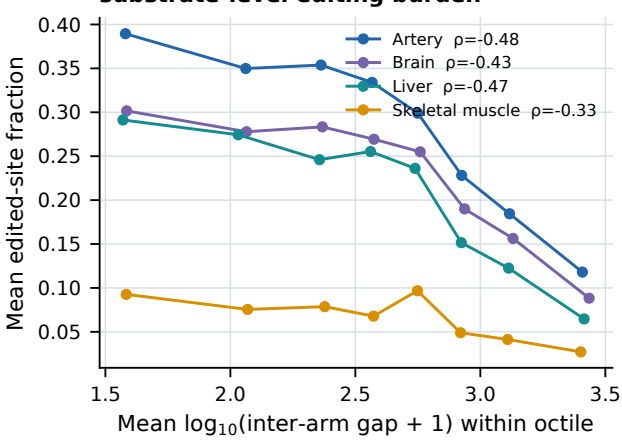
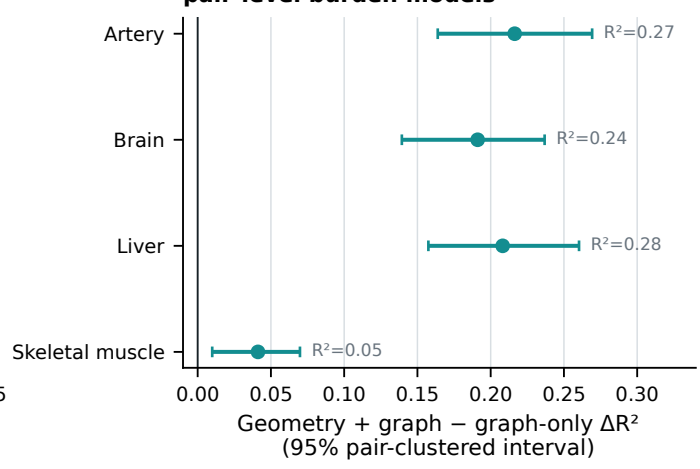
A

What adds to the sequence–structure graph?

Tested formulation	Evidence	Main finding
LEARNING & SUPERVISION		
Pair-first expanded supervision	Selected	More labels; correct pair weight
Continuous + ordinal auxiliary labels	Limited	Small ranking gain
Shared four-tissue training	Limited	Transfer below preset criteria
TARGET REPRESENTATION		
Aligned target + helix-partner view	Selected	Passes shifted-window control
Focal node / contact geometry	Limited	Localizes target effect
Frozen RNA-language embeddings	Not supported	F1-only; no ranking gain
Substrate-wide all-A decoder	Not supported	Stopped for futility
Larger / deeper graph networks	Not supported	No consistent benefit
Graph Transformer / GATv2 expansion	Not pursued	Simpler validated model sufficient
SUBSTRATE & CELLULAR CONTEXT		
Pair geometry / editing burden	Supported*	Substrate propensity detectable
Hierarchical engagement gate	Limited	Ranking gain; unstable F1
Bulk H3K36me3 overlap	Limited	Liver AUPRC only
RBP binding × bulk expression	Not supported	No residual gain
Transcript / repeat annotation panel	Not supported	No site-level gain
Matched donor/cell multimodality	Deferred	Matched data unavailable
Matched in vivo structure / isoforms	Deferred	Matched data unavailable
INTEGRATION		
Fixed whole-duplex + local-augmented logits	Selected	Complementary predictions
Unified end-to-end architecture	Not supported	Below frozen ensemble

See Table S2 for comparators, effect sizes, and intervals.
*Reproducible observation not yet useful for site prediction; deferred hypotheses were not tested.

B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